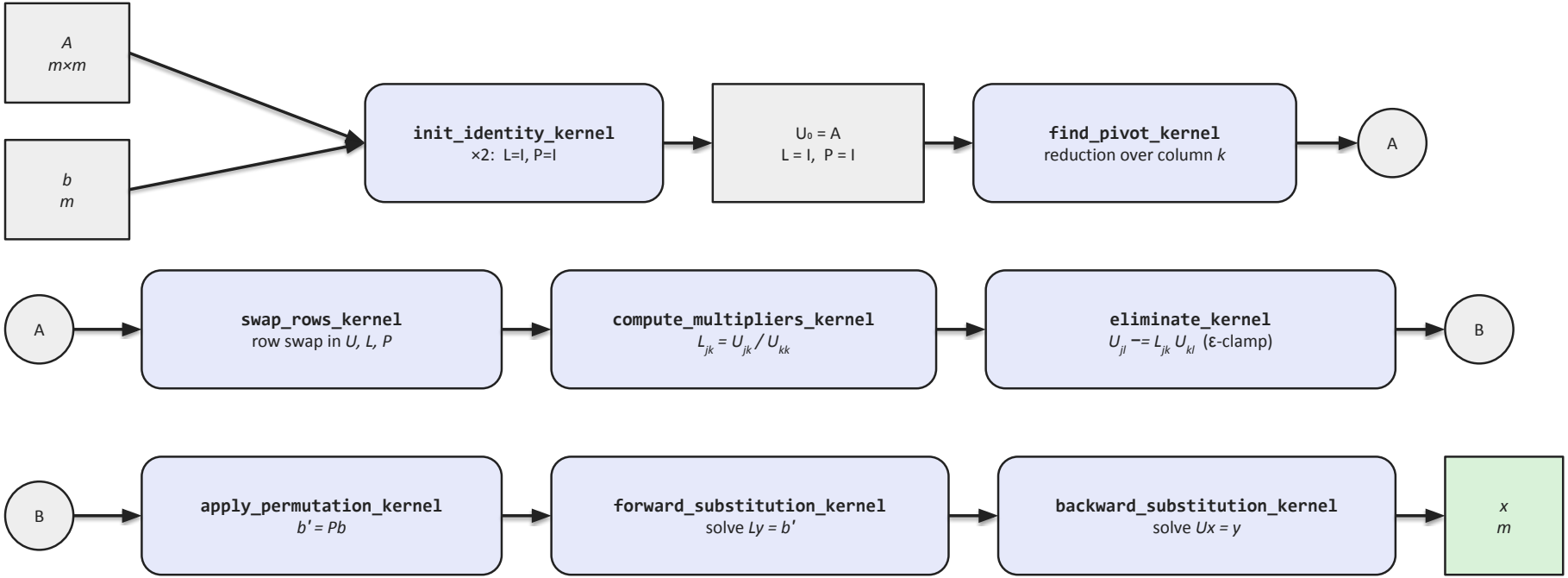
Algorithm: LU with partial pivoting, solve $Ax = b$

$PA = LU$ pivot: $p = argmax_{i \geq k} |U_{ik}|$ multiplier: $L_{jk} = U_{jk} / U_{kk}$
eliminate: $U_{jl} \mathrel{-}= L_{jk} U_{kl}$ $(j>k,\ l \geq k)$ forward: $Ly = Pb$ backward: $Ux = y$



NixNan Knobs: Environment Variables

INSTR_BEGIN / END	instr. index window
TOOL_VERBOSE	setup diagnostics
ENABLE_FUN_DETAIL	name on every rerun
PRINT_ILL_INSTR	print culprit SASS
SAMPLING	every N-th launch
INSTRUMENT_EXCEPTIONS	master switch
INSTR_MEM	also check STG (excn → mem)
HISTOGRAM	exponent histogram
BIN_SPEC_FILE	ranges to watch

NIXNAN_TIMEOUT	self-terminate, secs
LOGFILE	redirect output
LINE_INFO	source file:line (YMMV)
LOG_KERNELS	log launches only
MAX_ERRORS	stop after N reports
TIME_KERNELS	per-kernel time
FUNCTION_WHITELIST	only these kernels
FUNCTION_BLACKLIST	all but these

For example, ksel-autonixnan scripts (using the above env vars) the collection of kernel names and execution times, so users can then white- or black-list exactly what to observe in detail.

TIME_KERNELS: the first-launch cost

Kernel	Launches	First launch (µs)	Median later (µs)	Ratio
`init_identity_kernel`	2	312,451	21	≈ 14,900×
`find_pivot_kernel`	19	298,100	20	≈ 14,900×
`swap_rows_kernel`	19	290,205	25	≈ 11,600×
`compute_multipliers_kernel`	19	295,432	20	≈ 14,800×
`eliminate_kernel`	19	278,808	20	≈ 13,900×
`apply_permutation_kernel`	1	300,320	—	—
`forward_substitution_kernel`	1	326,736	—	—
`backward_substitution_kernel`	1	324,597	—	—

- First launch of every kernel: ≈280–330 ms. Later launches: ≈20–25 µs, four to five orders of magnitude less.
- The timer starts before NVBit rewrites the kernel's SASS (once per function) and stops after, so the first number is instrumentation cost, not kernel time.
- Kernels launched once (last three rows) only ever show this one-time cost.
- Library-heavy binaries pay far more: the cuSOLVER-based pytorch_slogdet run was killed at 120 s during instrumentation.

Stage 1 results: well-conditioned vs ill-conditioned input

	matrix.csv	bad_matrix.csv
Residual $\|\|Ax - b\|\|_2$ (uninstrumented)	9.68e-16	nan
Binade-threshold reports	99	63
FP64 NaN (repeats)	0 (0)	316 (2171)
FP64 Infinity (repeats)	0 (0)	4 (0)
FP64 Division-by-zero (repeats)	0 (0)	4 (4)

matrix.csv
clean solution, no exceptions

bad_matrix.csv
near-zero pivot → division by zero → NaN through the whole solve

Binary instrumentation reveals exceptions you did not expect

- The source computes only in double, yet bad_matrix.csv also reports 6 FP32 NaN detections (93 repeats).
- Cause: ptxas uses the idiom FFMA R8, RZ, R5, R3 (0 × R5 + R3 = R3) as a register move, shuffling the 32-bit halves of a double. nixnan sees an f32 FFMA whose operand R5 decodes as NaN.
- Lesson: a SASS-level detector reports what the compiled code does, not what the source says. A nonzero count in a format the source never uses is not necessarily that format's arithmetic.

